Innovation Prototype

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INTRODUCTION

Prototyping is a key factor in enhancing innovation capabilities. Research shows that prototyping increases hands-on collaboration and helps teams better visualize products, work iteratively, and incorporate varying ideas to build better products. (Berglund & Grimheden, 2011). In the increasingly globalized market, educational institutions embrace technological solutions to enhance student experiences and streamline administrative processes. This paper presents a prototype of our product "**HUSKY CONNECT**", which is a comprehensive student management system that integrates career development, event management, and financial planning functionalities.

PRODUCT DESCRIPTION

Husky Connect is tailored to address the unique challenges faced by international graduate students in the U.S. Combining career development tools, financial management resources, and cultural integration support, this one-stop solution offers a seamless experience accessible through mobile and web.

Key features include:

1. **Career Development**: Tools and resources for job hunting, including on-campus positions and internships. The platform offers resume workshops, webinars, and networking opportunities in collaboration with career service providers.
2. **Financial Management**: Budget calculators, personalized expense management tools, and financial literacy resources help students navigate monetary pressures.
3. **Cultural Integration through events**: Community-building events, cultural guides, and social networking opportunities foster a sense of belonging and help ease homesickness.

Designed for effortless adoption, the platform integrates with familiar systems like Canvas and Google Classroom. Its freemium model offers free access to essential features, with premium upgrades for personalized services.

LINKING PROTOTYPE FEATURES TO USER PROFILES

The prototype aligns closely with insights from the user profile and value proposition canvas.

1. **User Data Integration**
   Demographics, preferences, and behaviors informed the design, addressing challenges like career planning, networking, and budgeting. Features like the *"Hub"* and intuitive navigation were developed to resolve pain points.
2. **Value Proposition Fit**
   Design elements reflected core promises from the value proposition canvas. Pain relievers, such as centralized tools, and gain creators, like career and event planning features, directly addressed user goals.
3. **Prototype Testing Evidence**
   - **User Feedback:** Real users provided input on usability and satisfaction.
   - **Metrics and Analytics:** Data on feature use and task success informed improvements.
   - **Iterative Updates:** Adjustments like centralized navigation and visual enhancements improved alignment with user needs.
4. **Empathy-Driven Design**
   Real user data guided development, ensuring the prototype evolved as a solution tailored to user needs. Each feature reflects an understanding of their goals and frustrations, highlighting the commitment to a user-centric approach.

This approach ensured a functional, intuitive prototype that meets user expectations while aligning with the value proposition canvas.

UI/UX

The platform's design prioritizes modern UI/UX principles to create a seamless and engaging experience for international graduate students at Northeastern University. Key elements like the **login page**, **homepage**, intuitive **navigation**, and tools for **job search**, **events**, and **budgeting** are crafted for ease of use and accessibility. Using **Poppins** font enhances readability, while

responsive layouts ensure functionality across devices. Every feature emphasizes clarity and usability to align with students' needs.

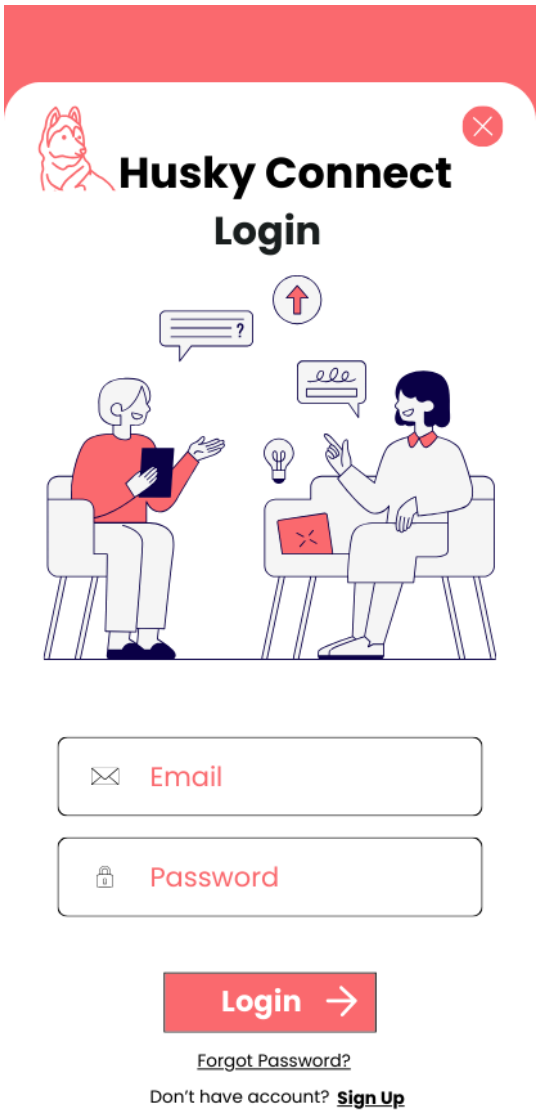
Branded with Northeastern's **pink and white** color palette and the iconic **Husky mascot**, the design fosters a sense of identity and belonging. Canva is employed to create cohesive visuals, ensuring consistency with the university's aesthetic. This combination of intuitive functionality and strong visual branding delivers a tailored platform that simplifies student tasks while reinforcing community connection.

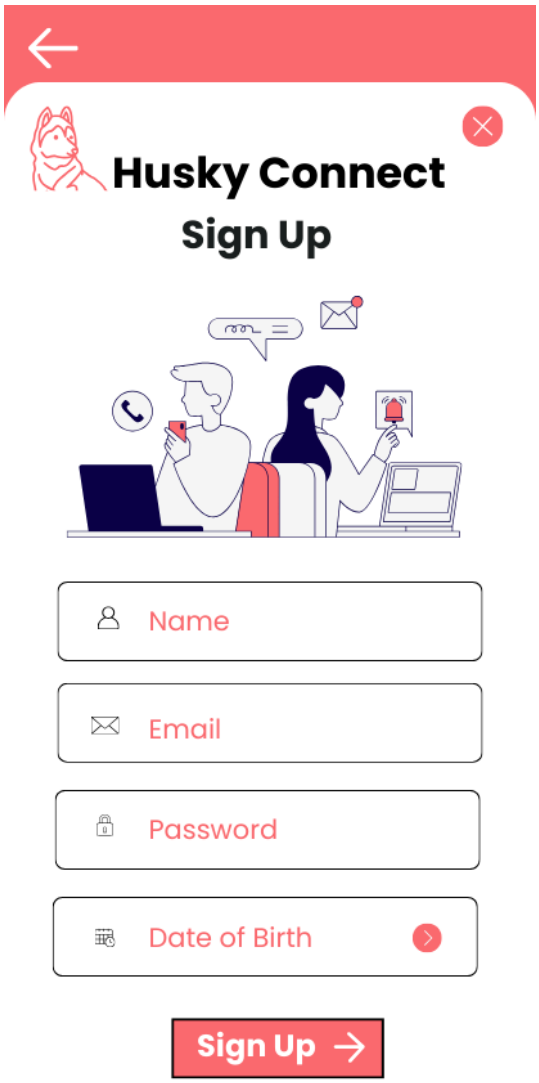
CANVA FOR WIREFRAMING

Canva is a powerful tool for wireframing, offering code-free visual editing and a user-friendly interface. Its drag-and-drop functionality and extensive library of templates make it easy to design layouts quickly, even without technical expertise. This simplifies creating intuitive wireframes for platforms focused on seamless UX/UI.

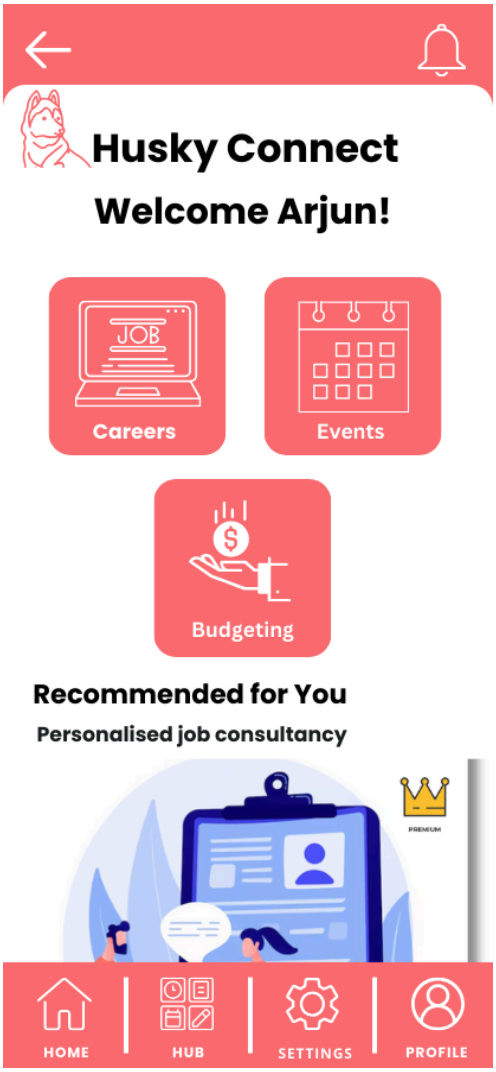
Collaboration is a key strength of Canva, with real-time commenting and editing features that streamline team feedback. Stakeholders can review, suggest changes, and refine designs efficiently, ensuring wireframes evolve to meet user needs. Canva's ease of use and collaboration tools make it ideal for crafting visually cohesive, user-focused prototypes. (Alli, 2022)

USER FLOW

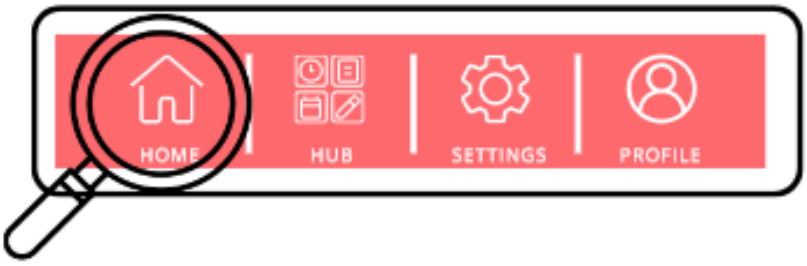




Login/Sign-up Page: The login page is a crucial entry point for users, especially international students with varying levels of technological proficiency, and should prioritize simplicity, security, and accessibility. Users will also have the option to create and account if they want to access the app. Following Park's (2022) insights on UX/UI branding, the design should be clean and intuitive, with clear instructions and error messages to minimize cognitive load. Additionally, implementing features such as password recovery, and remember-me functionality, as recommended by Xavier et al. (2023), ensures both user satisfaction and robust security, enhancing the overall user experience.

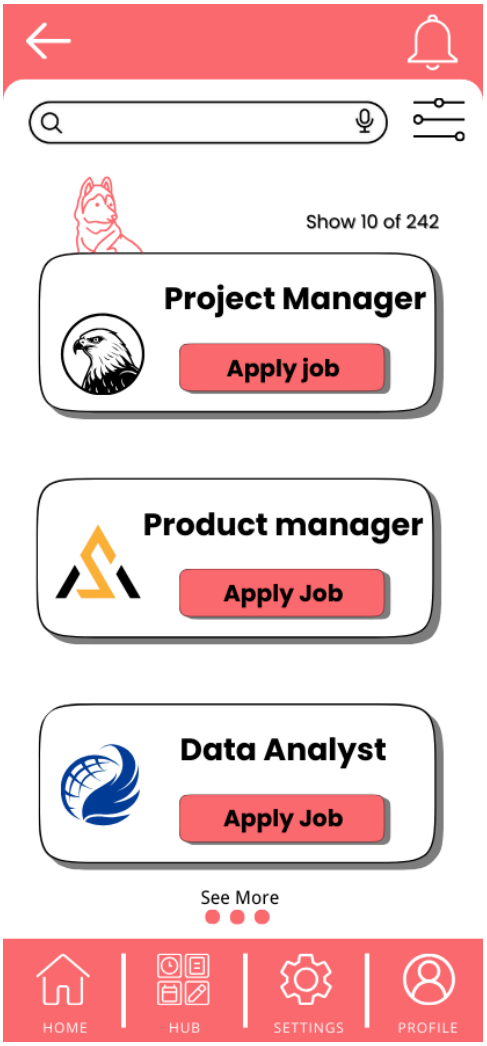


Homepage: The homepage acts as the gateway to the platform, providing a clear and intuitive overview of key features such as job search, events, and budgeting tools, ensuring international students can easily find relevant information. Dosi et al. (2020) highlight that prototypes serve as identity markers, and the homepage, reflecting the platform's purpose and ethos, should engage users from the start. In line with Xavier et al.'s (2023) findings on UI/UX satisfaction, the dashboard incorporates features like notifications, reminders, and interactive elements to enhance user engagement.



Navigation Bar: The Navigation bar is uniform across all screens of the application and offers a way to easily navigate to various parts of the app while not on the homepage.

- The home option helps users to navigate back to the homepage from anywhere on the application.
- The hub tab offers the users an option to view all the features and tabs on the app in one place.
- The profile feature helps users set up their app profile but also to be able to easily search, track applications, and access personalized content, offering a clear overview of saved jobs, application statuses, and upcoming events.
- The settings options are offered to users to set up their preferences for the app including customizable homepage widgets, notification preferences, and privacy settings.



Job Search

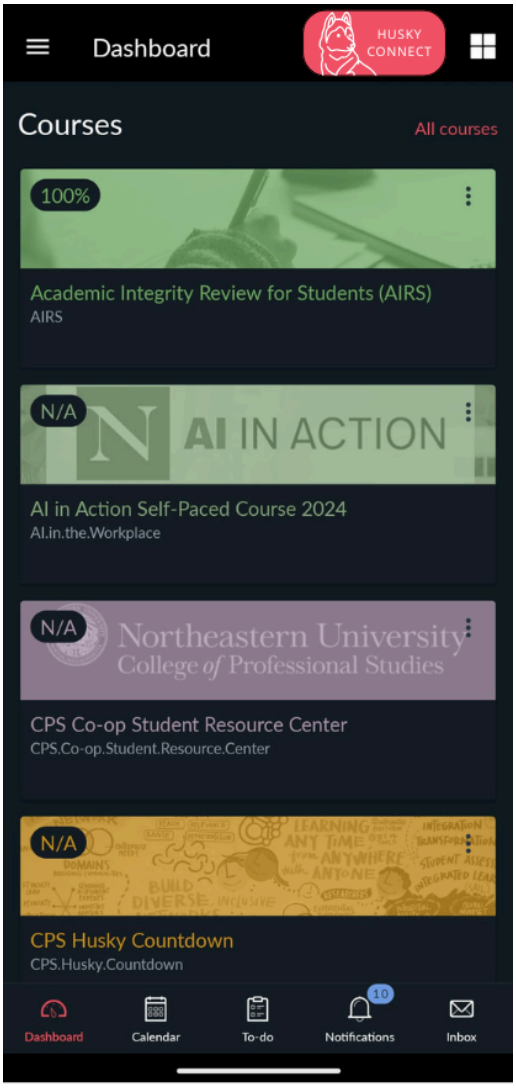
The job search functionality is central to the platform, enabling international students to explore job listings tailored to their qualifications and locations. Designed with Dosi et al.'s (2020) principles in mind, it features advanced filtering options, a search bar, top company profiles, and an easily accessible apply button. The interface prioritizes user experience, ensuring fast, responsive performance across devices, and handles large volumes of job listings seamlessly. Following Park's (2022) UX/UI design guidelines, the platform includes personalized job alerts and recommendations to enhance efficiency, helping students navigate the job search process with ease and clarity.

Events

The event tab is a key feature for international students, providing a calendar or listing of career fairs, industry seminars, and networking opportunities. Designed for simplicity and clarity, it allows students to RSVP, search for events, add events to their calendars, and receive reminders. Key event details, such as date, time, location, and registration links, are displayed, with special emphasis on events offering sponsorships or visa-related opportunities. Following Park's (2022) branding strategy principles, the event management section integrates a calendar, an event registration system, reminder notifications, and event categorization. Personalized recommendations and filters further assist students in making informed decisions about which events to attend.

Student Budget Planning

The student budget planning tool is an essential feature for international students, helping them manage financial challenges while studying abroad. Designed with simplicity and clarity, it includes pre-set budget categories such as housing, transportation, and groceries, along with customizable options for individual needs. Interactive graphs, progress trackers, and spending analytics encourage students to stay on top of their finances. Following Xavier et al.'s (2023) user-centric design principles, the tool offers features like expense tracking, budget visualization, and financial goal setting to enhance user engagement and support students in managing both their job search and financial responsibilities effectively.



Integration with Canvas

The integration of a dedicated tab on the Canvas dashboard provides users with seamless access to our platform. Located in the upper-right corner, the pink Husky button and logo serve as a distinct and intuitive entry point, ensuring brand visibility and easy navigation. This design not only reinforces our identity with the Northeastern community but also simplifies the user journey by embedding the app directly within a familiar interface.

PROTOTYPE EVOLUTION BY DATA ANALYSIS

1. Initial Concept and Ideation: The foundation of the prototype is derived from the user profile analysis and value proposition canvas. Key user pain points and gain creators are identified as the driving forces behind the initial design to incorporate Budget, Events, and Career options on the homepage. We aimed to keep the wireframe clean and simple. 100% of the tested users needed a one-point solution for jobs, events, and budgeting

2. First Prototype and Testing: A low-fidelity prototype was created, focusing on essential features for usability testing. We started with an all-pink base, however, 70% of the users preferred a white base with pink accents for the prototype since it was easier to navigate.

3. Iterative Refinements: The prototype was changed iteratively. An example of incorporating user feedback would be that 60% of the audience wanted all resources of the app in one place, hence the “Hub” icon was created. Streamlining navigation based on heatmap data. We also Added features requested by users, such as enhanced accessibility or additional customization such as the search bar and filter options which were requested by 23% of the users

4. Final Prototype: The final iteration of the prototype is showcased, demonstrating its polished design and comprehensive functionality. Each feature is tied back to the original user insights and value proposition canvas, emphasizing that the end product meets the target audience’s needs.

CONCLUSION

This comprehensive student management system for international students is designed with a user-first approach, prioritizing accessibility, ease of use, and functionality. By integrating key features such as a personalized dashboard, job search functionality, event tracking, and student budget planning, the platform would provide a comprehensive resource to help students navigate the challenges of job hunting and financial management in a foreign country. The application of sound UX/UI principles, as outlined by Park (2022) and Xavier et al. (2023), ensures a seamless user experience, fostering continued engagement and satisfaction. This platform empowers

international students with the tools needed for success and exemplifies a significant advancement in educational technology integration. The system effectively addresses the diverse needs of today's student population while maintaining high usability standards. Ultimately, the platform serves as a vital tool in supporting international students' academic and professional journeys.

This prototype effectively adds value to international students' lives by offering a one-stop solution for career development, financial management, and social integration. By prioritizing user experience, offering customizable resources, and integrating with essential academic platforms, it creates a seamless, supportive environment for international students to thrive. The platform's comprehensive approach not only addresses the immediate challenges faced by these students but also provides long-term benefits by fostering personal growth, academic success, and professional development.

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